

GEBZE TECHNICAL UNIVERSITY
ELECTRONIC ENGINEERING
ELC 336

Name	Fouad	Büşra
Surname	Aladhami	Ülker
Student ID	210102002808	200102002011

Performance Table

GAIN[dB]	40.2211
BW[MHz]	1.0797
PH[Degree]	53.2200
POWER[mW]	0.7377

FOM=3132.9404

Table of contents

1 INTRODUCTION 4
2 ANALYTICAL SOLUTION & SIMULATION 4
3 CONCLUSION 10
4 APPENDICES 10
5 REFERENCES 12

1 INTRODUCTION

In this project, it is desired to design an active loaded differential amplifier circuit to drive a 0.5 pF capacitor while respecting some limits. Current is supplied to the circuit using mirror current. Gain of 100[V/V], 3dB BW of 1MHz, and Phase Margin of 50-degree, power consumption 1mW are selected and the design is started.

2 ANALYTICAL SOLUTION & SIMULATION

Firstly[1], the circuit was set up for active loaded differential amplifier with pnp transistors as inputs and NPN transistors as active load. Then the current source was added to the circuit by selecting PNP according to the circuit.

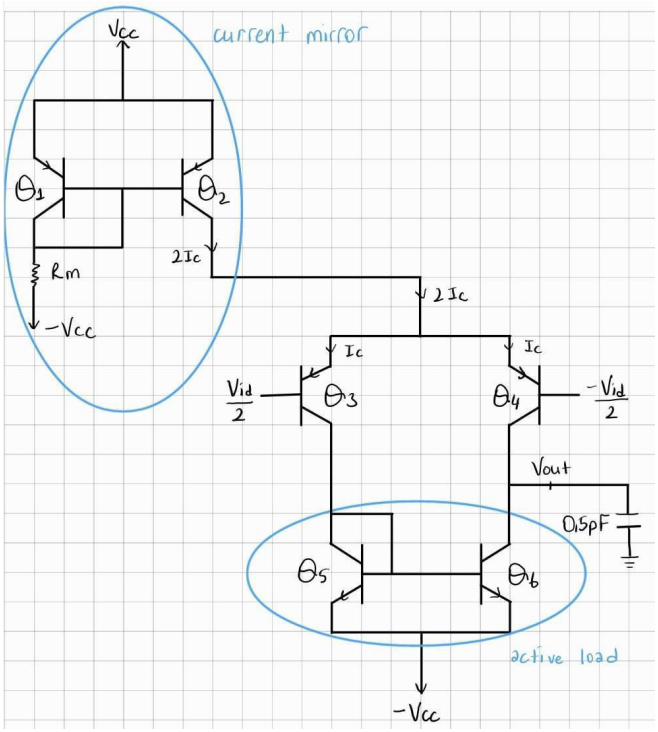


Figure 1:First Circuit.

Then the current source was adjusted by considering the power consumption. Since the current has a positive effect on the bandwidth, the voltage was kept at a low level and the calculation was made.

To ensure power consumption:

$$Vcc = 1V \quad -Vcc = -1V$$

$$\text{And } P = 8 * Ic * 1V = 10^{-3}mW$$

$$\text{So } Ic = 0.125\,mA \text{ and } Rm = 5.5\,kohm$$

$$Acircuit(gain) = gm4 * (ro4//ro5) = gm4 * \frac{\frac{Va4 * Va5}{Ic^2}}{\frac{Va4 + Va5}{Ic}}$$

$$gm4 = \frac{Ic}{Vt} = \frac{0.125 * 10^{-3}}{25 * 10^{-3}} = 5\,mS$$

$$ft = \frac{gm4}{2 * pi * Ctotal} = 10^8 Hz$$

$$Ctotal = \frac{gm4}{ft * 2 * pi} = 7.957 * 10^{-12}$$

While the gain is achieved by keeping the Va value high, the desired BW was tried to be achieved by keeping the base collector capacity value of the transistor low and the best one was found with the help of python code(The code is provided in appendices section) , but the desired BW could not be obtained as seen in Figure 2 and Figure 3 because of the high gain. And reduce the gain to the desired level is not possible with the because the transistors that have low Va , have a high Base Emitter capacitance that cause a low BW.

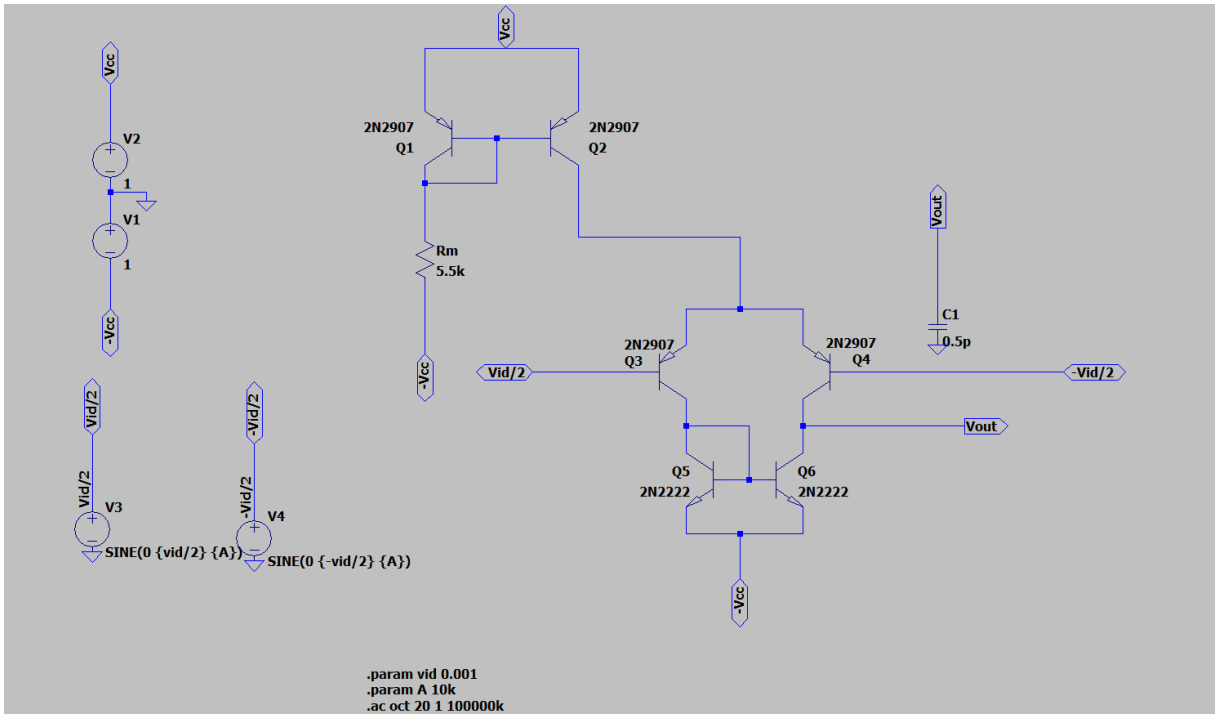


Figure 2: First design circuit.

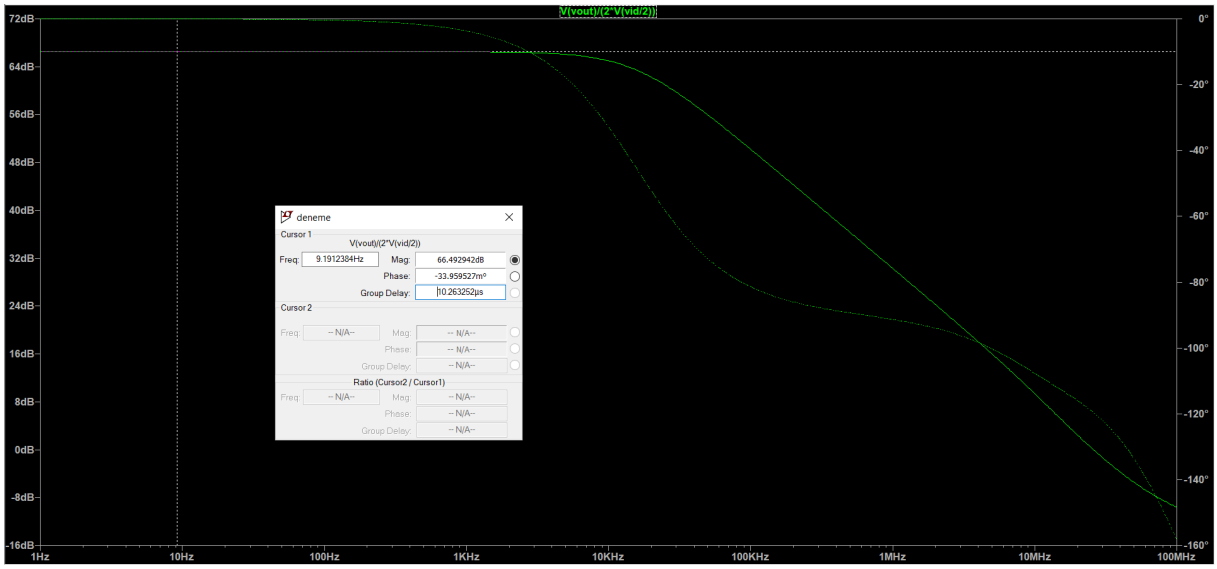


Figure 3: First design circuit AC analysis.

Since the gain value is more than the limit and the BW is narrow, it is desired to expand the BW by decreasing the gain to increase BW with the feedback circuit.

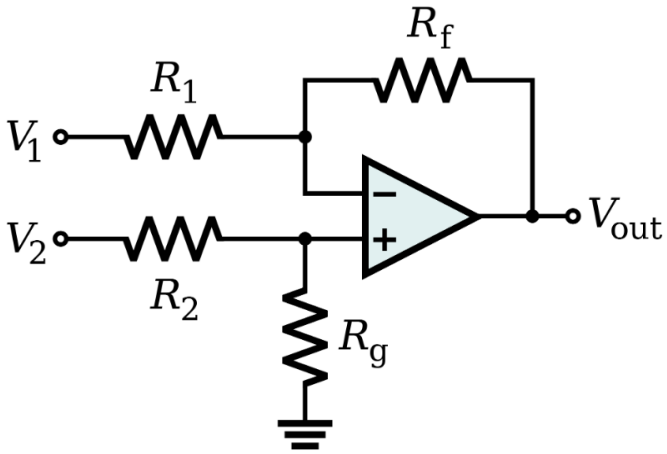


Figure 4: Feedback for differential amplifier.

$$When\ V1\ = \left(\frac{Vid}{2}\right) and\ V2\ = \left(-\frac{Vid}{2}\right)$$

$$Af\ = \frac{Rf}{R1}(for\ R1 = R2, Rf\ = Rg)$$

When the feedback circuit in Figure 4 was added to the system, it was observed that the gain decreased, and the BW decreased as expected. However, since the open loop gain of the OPAMP used in the calculation is very high (near infinity), approximately 100[V/V] gain was obtained in the real circuit instead of 560[V/V] gain obtained in the equation for ideal OPAMP. The values of feedback resistances are funded by soft tuning the resistor values. And the funded circuit is shown in the Figure 5.

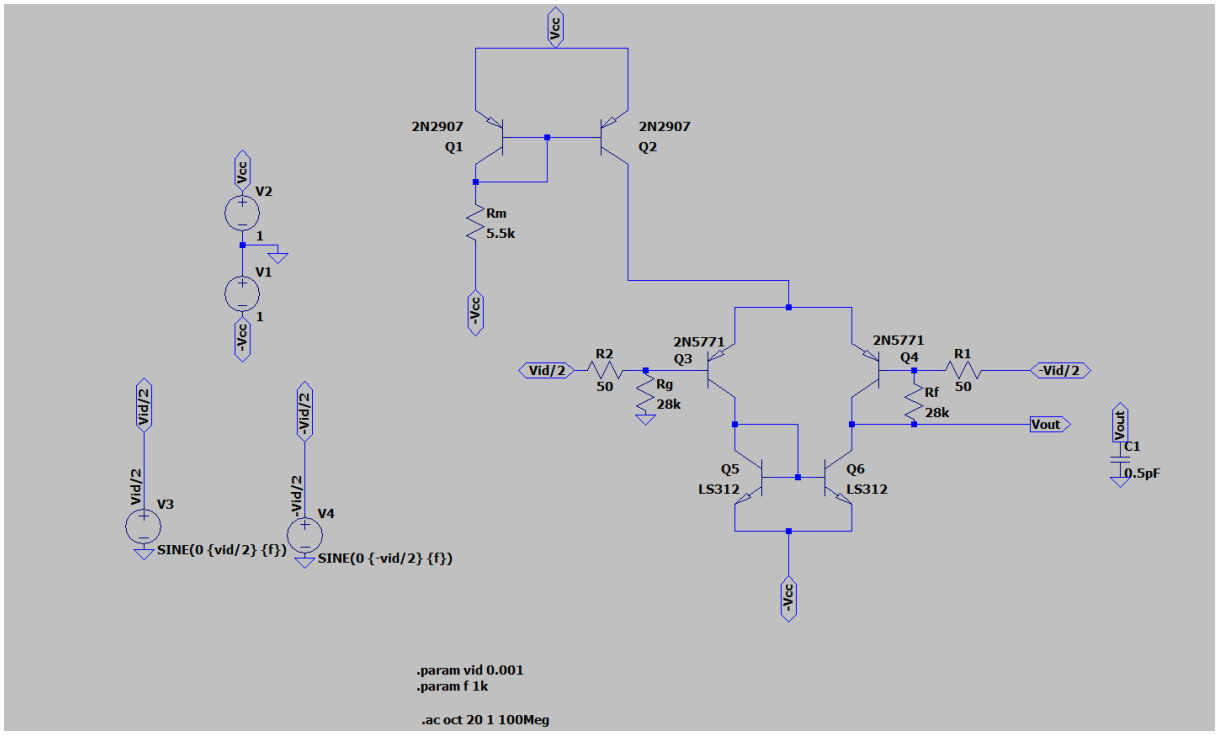


Figure 5: Differential amplifier with feedback.

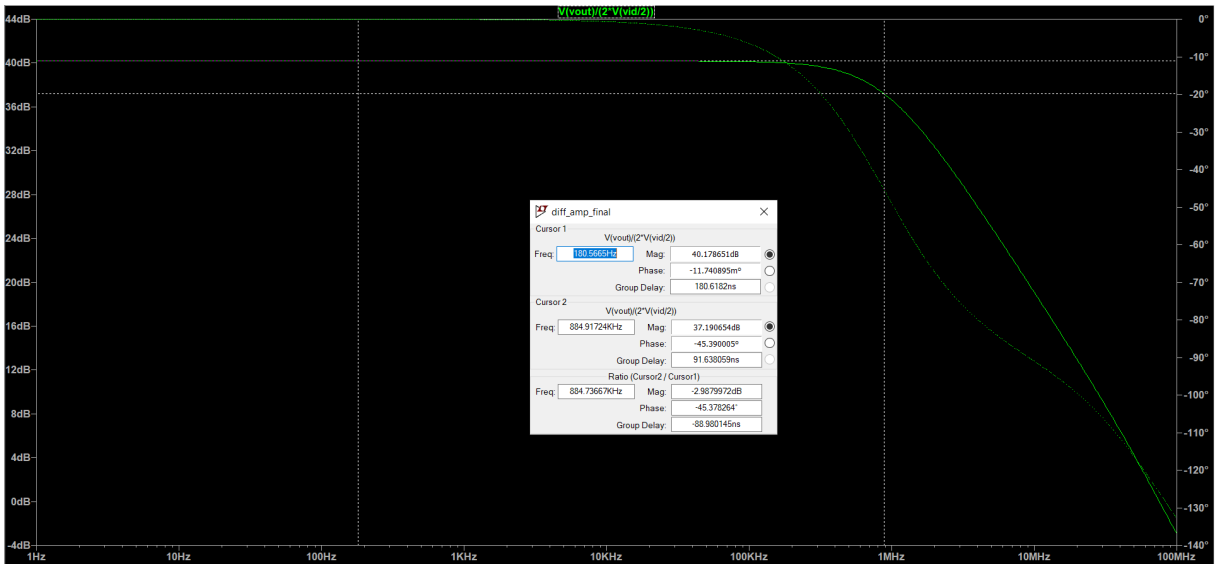


Figure 6: Differential amplifier with feedback circuit gain and BW.

As shown in Figure 6 a 44 dB gain obtained with a bandwidth near 900Khz but it's not enough. Gain can be reduced to increase bandwidth by tuning the feedback. So, if the gain reduced to 40 dB a 1MHZ bandwidth obtained.

To maximize the FOM minimizing the power is possible. As seen in Figure 5 the current mirror is designed with 2 PNP transistors. So, the current across R_m, I_{Rm} is copied, and the amplifier circuit is biased with I_{Rm}. So, to bias the amplifier circuit the 50% of power transferred by the DC feeding is loss across the resistance R_m.

Then, as shown in Figure 7, the current supplied to the circuit was increased without changing the voltage value by taking advantage of the parallelism of transistors in current mirror circuit. At this circuit current across R_m is I_{Rm} so the amplifier circuit is biased by current near 7 I_{Rm} because of the parallel 7 transistors. So, the loss of power is decreased from 50% to (1/8) = 12.5%. And the amplifier bias current increase. It's a big advantage, because if the power loss decreases, the amplifier circuit can be biased by the same current but by using less power.

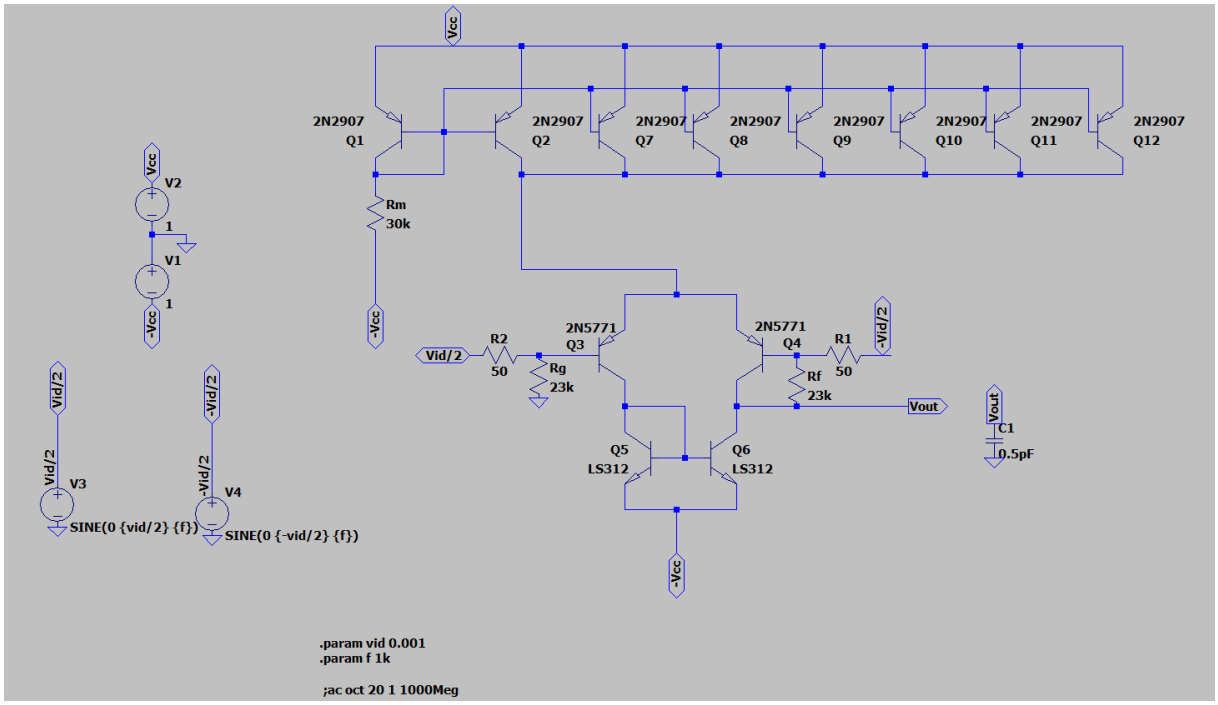


Figure 7: Final Differential amplifier circuit.

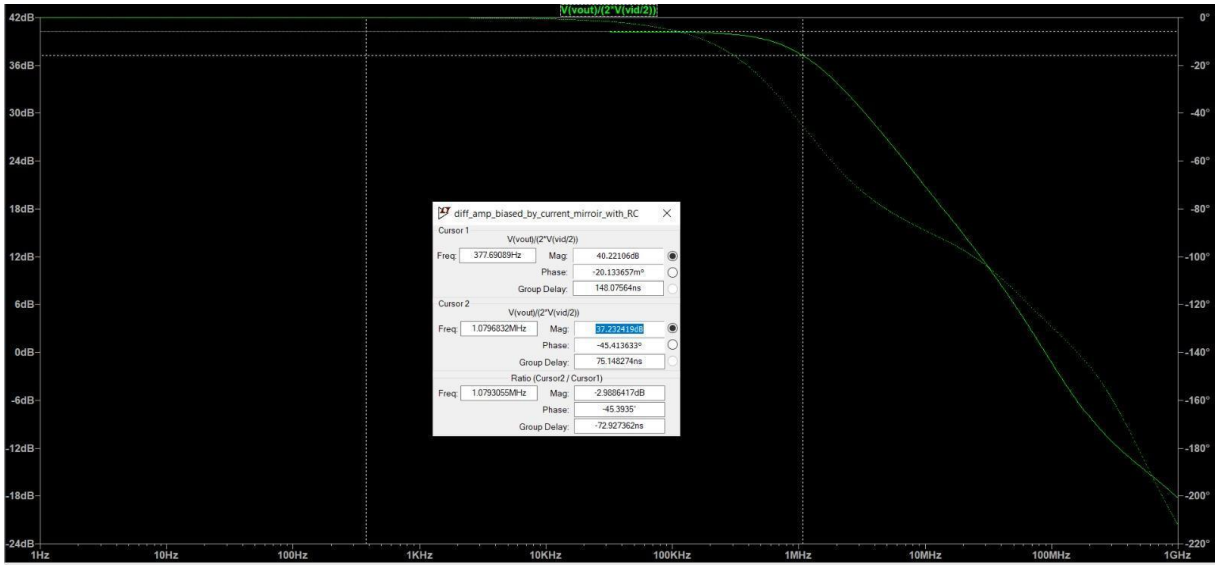


Figure 8: Final Differential amplifier circuit gain and BW.

As shown in Figure 8, after reducing the power of the circuit and retune the feedback circuit. A 40.22 dB gain is obtained with a bandwidth 1.0796Mhz. So the circuit meat the basic Gain and BW requirements.

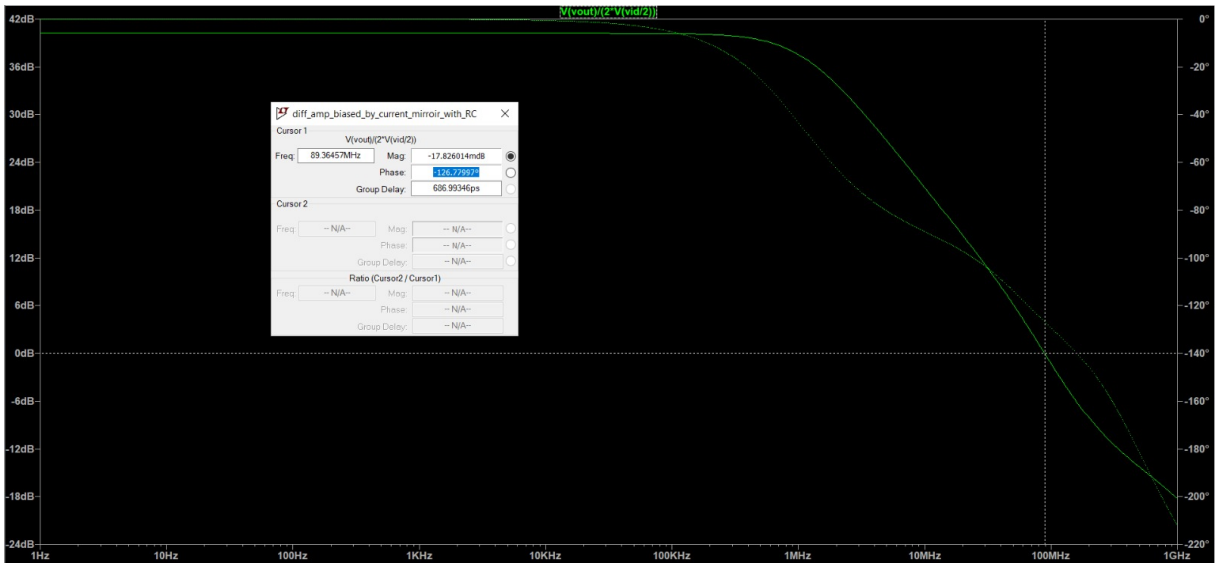


Figure 9: Final Differential amplifier circuit phase margin.

The phase margin of the amplifier is defined as the following:

$$PM° = (\,Phase\ at\ unity\ frequency)° + 180$$

And as shown in Figure 9 the phase angle at unity frequency its equal to -126.77 so

$$PM° = 53.22°$$

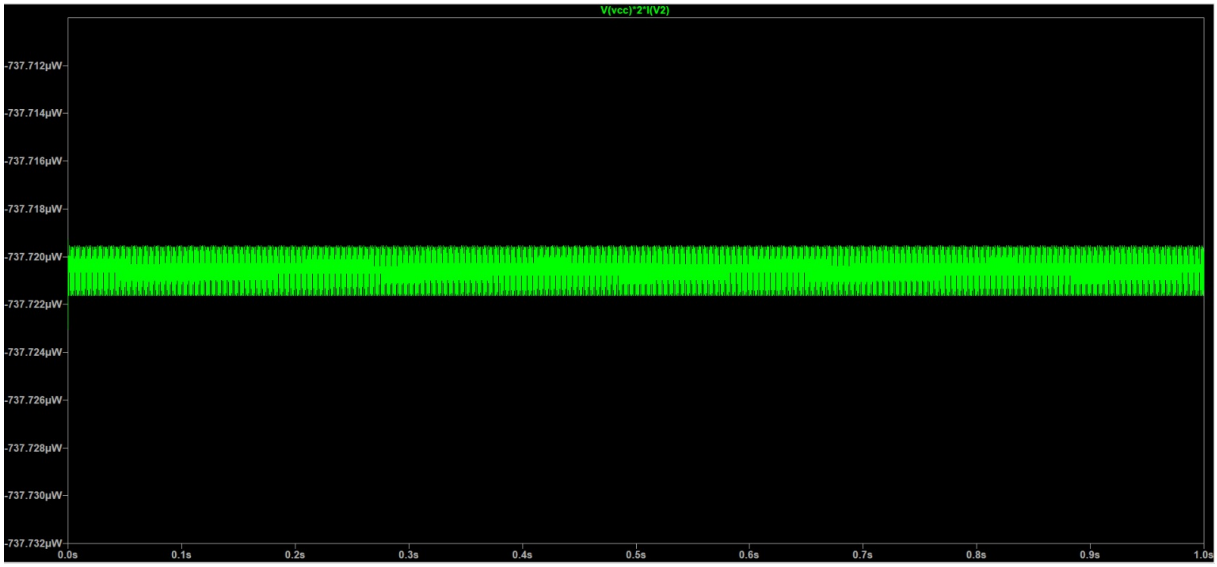


Figure 9: Final Differential amplifier circuit power consumption.

To show the power consumptions of the circuit of the circuit a transient analyze is make in mid band frequency and the power is showed in Figure 9.

3 CONCLUSION

As a result, an active loaded differential amplifier circuit has been constructed and information about feeding the circuit with a current mirror has been obtained. Then, by limiting the power, the gm value was calculated, and the ft value was determined, and the limits were tried to be provided by looking at the Vaf and capacitor values of the transistors. However, when the desired limits could not be fully achieved, the inverse relationship between gain and BW was utilized and the high gain was reduced with the feedback circuit and the requirements were obtained approximately. Then, to improve the circuit, the current was increased with parallel transistors attached to the current mirror and the power consumed was reduced by increasing Rm. In this project, it has been learnt how to follow a path in circuit design and the effect of limitations on circuits.

4 Appendices

The python script to find the transistor with lower value than the provided Va and Cjc is provided as follow. The LTspice built in library “standart.bjt” is used to find the transistors.

```
import re

# Function to parse a transistor model line
def parse_transistor_line(line):
    parts = line.split()
    model_name = parts[1]
    model_type = parts[2]
    parameters = ' '.join(parts[3:])
    return model_name, model_type, parameters
# Function to convert values with 'p' (picofarad) to numerical values
def convert_to_numerical(value):
    if 'p' in value.lower():
        return float(value.lower().replace('p', '')) * 1e-12
    elif 'f' in value.lower():
        return float(value.lower().replace('f', '')) * 1e-15
    elif 'n' in value.lower():
        return float(value.lower().replace('n', '')) * 1e-9
    elif 'u' in value.lower():
        return float(value.lower().replace('u', '')) * 1e-6
    elif 'm' in value.lower():
        return float(value.lower().replace('m', '')) * 1e-3
    elif 'k' in value.lower():
        return float(value.lower().replace('k', '')) * 1e3
    elif 'meg' in value.lower():
        return float(value.lower().replace('meg', '')) * 1e6
    elif 'g' in value.lower():
        return float(value.lower().replace('g', '')) * 1e9
    elif 't' in value.lower():
        return float(value.lower().replace('t', '')) * 1e12
    else:
        return float(value)
# Function to extract VAF and CJE values from the parameters
def extract_vaf_cje_numerical(parameters):
```

```
    vaf_match = re.search(r'Vaf=(\d+\.?\d*(?:[eE][-+]?\d+)?[pP]?)',
parameters, re.IGNORECASE)
    cje_match = re.search(r'Cjc=(\d+\.?\d*(?:[eE][-+]?\d+)?[pP]?)',
parameters, re.IGNORECASE)

    vaf_value = float(convert_to_numerical(vaf_match.group(1))) if
vaf_match else None
    cje_value = float(convert_to_numerical(cje_match.group(1))) if
cje_match else None
    return vaf_value, cje_value

# Function to check if the transistor meets the specified criteria
def does_meet_criteria_numerical(parameters, vaf_limit, cje_limit):
    vaf_value, cje_value = extract_vaf_cje_numerical(parameters)
    return (vaf_value is not None and vaf_value < vaf_limit) and
(cje_value is not None and cje_value < cje_limit)

# Function to concatenate multi-line model definitions into single
lines
def concatenate_multiline_models(file_lines):
    concatenated_models = []
    current_model = ""
    for line in file_lines:
        if line.startswith('.model'):
            if current_model:
                concatenated_models.append(current_model)
            current_model = line.strip()
        elif line.startswith('+'):
            current_model += ' ' + line.strip()
    else:
        if current_model:
            concatenated_models.append(current_model)
    return concatenated_models

# Vaf and CJE limits (You can adjust these values)
vaf_limit = 1000  # Example limit for Vaf
cje_limit = 5e-12  # Example limit for CJE

# Reading the file and processing the transistors
file_path =  'standard.bjt'  # Replace with your file path
with open(file_path, 'r', encoding='ISO-8859-1') as file:
    file_content = file.readlines()

concatenated_models = concatenate_multiline_models(file_content)
transistors_meeting_criteria = []

for model_line in concatenated_models:
    model_name, model_type, parameters =
parse_transistor_line(model_line)
    if does_meet_criteria_numerical(parameters, vaf_limit,
cje_limit):
        vaf_value, cje_value = extract_vaf_cje_numerical(parameters)
        if vaf_value is not None and cje_value is not None:
            transistors_meeting_criteria.append((model_name,
model_type, vaf_value, cje_value))

transistorType = "npn"
```

```
# Printing the filtered transistors
for transistor in transistors_meeting_criteria:
    if transistorType in  transistor[1].lower() :
        print(transistor)
```

5 REFERENCES

[1] Adel S. Sedra and Kenneth C. Smith, Microelectronic Circuits,2015,SEVENTH EDITION